

CS 421 --- LL Parsing Activity

Name	Netid

Please write your name/netid legibly in dark ink. Hand in one copy per team. Do not staple or mangle the corners.

Purpose

An LR parser is a more flexible kind of parser than the LL parsers we covered before. In this activity you will learn how to use and generate LR tables.

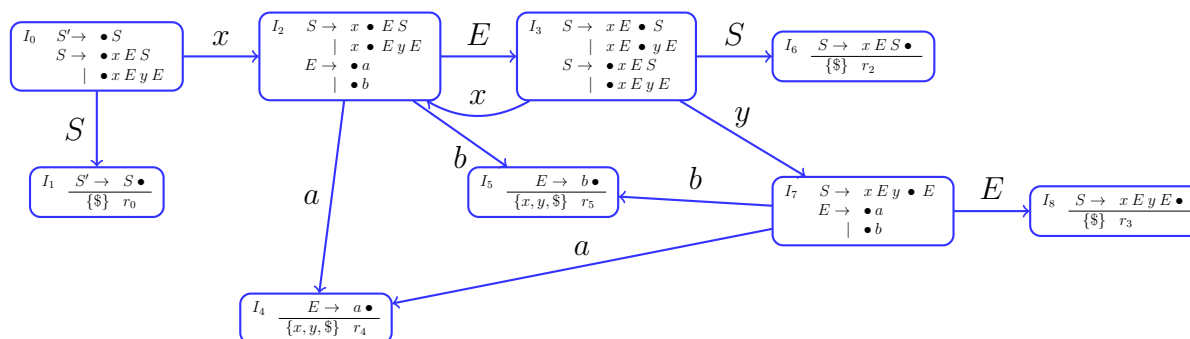
Your objectives:

- Use an LR table to parse a sentence.
- Create a state machine and LR table for a given grammar.
- Detect an ambiguity in a grammar.

Using the State Machine

Consider the following grammar and corresponding state machine.

$$\begin{aligned}
 S' &\rightarrow S \\
 S &\rightarrow x E S \\
 &\quad | x E y E \\
 E &\rightarrow a \\
 &\quad | b
 \end{aligned}$$



If we try to parse the sentence xaxbya, we get the following behavior. (See next page.)

Step	State	Input	Stack	Action
0	0	xaxbya		Shift x
1	2	axbya	(0,x)	Shift a
2	4	xbya	(0,x), (2,a)	Reduce $E \rightarrow a$
3	2	xbya	(0,x)	Goto 3
4	3	xbya	(0,x),(2, $E \rightarrow a$)	Shift x
5	2	bya	(0,x),(2, $E \rightarrow a$),(3,x)	Shift b
6	5	ya	(0,x),(2, $E \rightarrow a$),(3,x),(2,b)	Reduce $E \rightarrow b$
7	2	ya	(0,x),(2, $E \rightarrow a$),(3,x)	Goto 3
8	3	ya	(0,x),(2, $E \rightarrow a$),(3,x),(2, $E \rightarrow b$)	Shift y
9	7	a	(0,x),(2, $E \rightarrow a$),(3,x),(2, $E \rightarrow b$),(3,y)	Shift a
10	4	a	(0,x),(2, $E \rightarrow a$),(3,x),(2, $E \rightarrow b$),(3,y),(7,a)	Reduce $E \rightarrow a$
11	7		(0,x),(2, $E \rightarrow a$),(3,x),(2, $E \rightarrow b$),(3,y)	Goto 8
12	8		(0,x),(2, $E \rightarrow a$),(3,x),(2, $E \rightarrow b$),(3,y),(7, $E \rightarrow a$)	Reduce $S \rightarrow x E y E$
13	3		(0,x),(2, $E \rightarrow a$)	Goto 6
14	6		(0,x),(2, $E \rightarrow a$),(3, $S \rightarrow \dots$)	Reduce $S \rightarrow x E S$
15	0			Goto 1
16	0		(0, $S \rightarrow \dots$)	Accept

Problem 1) Draw the tree that resulted from this parse.

Problem 2) What is the purpose of the shift operation?

Problem 3) What is the purpose of the reduce operation?

Problem 4) In step 12, how does the parser know how many states to pop off the stack when doing the reduce?

Generating State Machines

Example 1)

$$\begin{array}{c} S \rightarrow x S y \\ | \quad \epsilon \end{array}$$

Action

Go To

	x	y	\$
0			
1			
2			
3			

	x	y	\$	S
0				
1				
2				
3				

Example 2)
$$\begin{array}{lcl} S & \rightarrow & a E b \\ & | & x \\ E & \rightarrow & E x E \\ & | & b \end{array}$$

Action

Go To

	a	b	x	\$
0				
1				
2				
3				
4				
5				
6				

	a	b	x	\$	S	E
0						
1						
2						
3						
4						
5						
6						

Problem 5)

Is this grammar ambiguous? Prove your answer by generating the LR parsing table.

$$\begin{array}{l} S \rightarrow v \\ \quad | \quad x S \\ \quad | \quad x S y S \end{array}$$

Action

Go To

	v	x	y	\$
0				
1				
2				
3				
4				
5				

	v	x	y	\$	S
0					
1					
2					
3					
4					
5					

Problem 6)

You will discover a new kind of conflict in this grammar.

$$\begin{array}{l} S \rightarrow a b \\ \quad | a E \\ \quad | F b \\ E \rightarrow b \\ F \rightarrow a \end{array}$$

Action

	a	b	c	\$
0				
1				
2				
3				
4				
5				

Go To

	a	b	c	\$	S	E	F
0							
1							
2							
3							
4							
5							